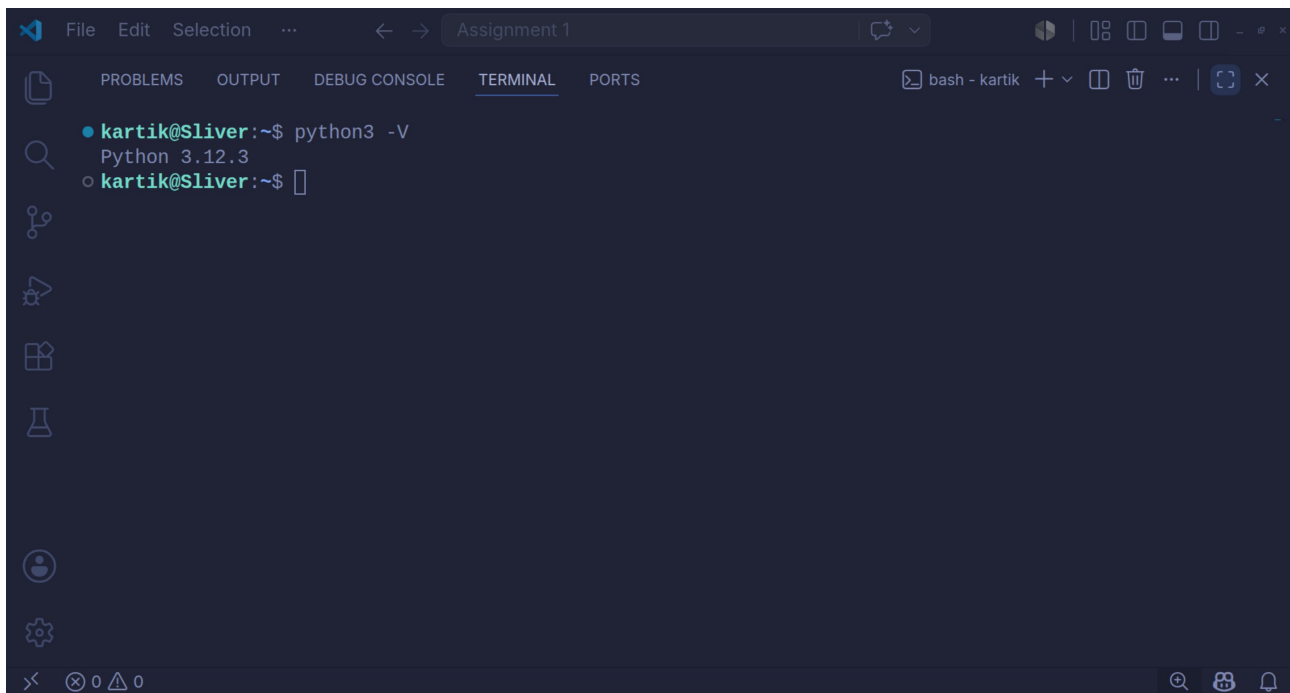


1. Install VS Code and Python

A screenshot of the Visual Studio Code (VS Code) interface. The top menu bar shows 'File', 'Edit', 'Selection', and a dropdown arrow. Below the menu bar, the 'TERMINAL' tab is active, displaying a bash shell session. The terminal output shows the command 'python3 -V' being executed, resulting in 'Python 3.12.3'. The prompt 'kartik@Sliver:~\$' is visible. The left sidebar contains icons for Explorer, Search, Source Control, Run and Debug, Extensions, Testing, and Settings. The bottom status bar shows '0' errors, '0' warnings, and '0' info messages.

```
File Edit Selection ... Assignment 1
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● kartik@Sliver:~$ python3 -V
Python 3.12.3
○ kartik@Sliver:~$
```

2. Write Pseudocode

a. Input a Number and Print Its Square

```
START
  INPUT number
  square = number × number
  PRINT square
END
```

b. Calculate the Average of Given Numbers

```
START
  INPUT n          // Number of values
  sum = 0

  FOR i = 1 TO n DO
    INPUT value
    sum = sum + value
  END FOR

  average = sum / n

  PRINT average
END
```

c. Multiply Two Integers

START

INPUT integer1

INPUT integer2

product = integer1 × integer2

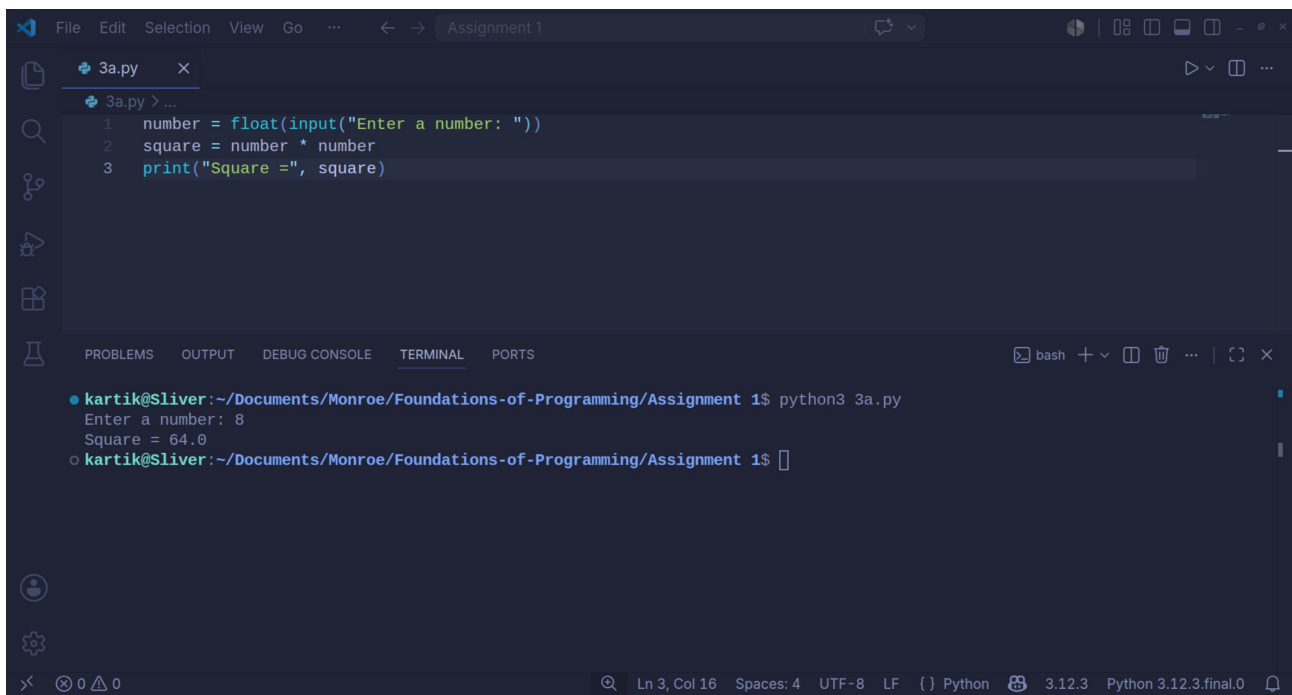
PRINT product

END

3. Write and Execute Python Programs

a. Input a Number and Print Its Square

```
number = float(input("Enter a number: "))
square = number * number
print("Square =", square)
```



The screenshot shows a Visual Studio Code editor window with a file named '3a.py' open. The code in the editor is:

```
1 number = float(input("Enter a number: "))
2 square = number * number
3 print("Square =", square)
```

Below the editor is a terminal window. The terminal shows the command to run the script and its output:

```
kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$ python3 3a.py
Enter a number: 8
Square = 64.0
kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$
```

The status bar at the bottom indicates the file is 'Ln 3, Col 16', uses 'Spaces: 4', 'UTF-8' encoding, 'LF' line endings, and is a Python file using 'Python 3.12.3'.

b. Calculate the Average of Given Numbers

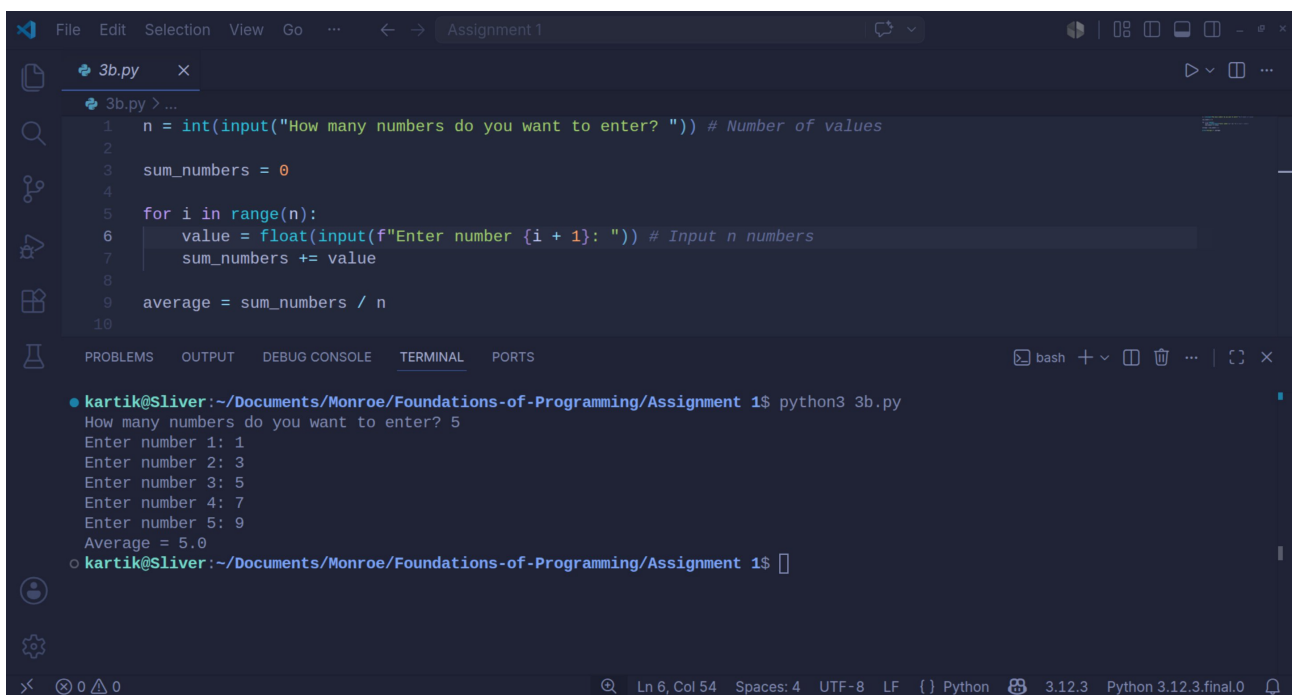
```
n = int(input("How many numbers do you want to enter? ")) # Number of values

sum_numbers = 0

for i in range(n):
    value = float(input(f"Enter number {i + 1}: ")) # Input n numbers
    sum_numbers += value

average = sum_numbers / n

print("Average =", average)
```



The screenshot shows a code editor with a file named `3b.py` containing the following Python code:

```
1 n = int(input("How many numbers do you want to enter? ")) # Number of values
2
3 sum_numbers = 0
4
5 for i in range(n):
6     value = float(input(f"Enter number {i + 1}: ")) # Input n numbers
7     sum_numbers += value
8
9 average = sum_numbers / n
10
```

Below the code editor is a terminal window showing the execution of the script:

```
● kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$ python3 3b.py
How many numbers do you want to enter? 5
Enter number 1: 1
Enter number 2: 3
Enter number 3: 5
Enter number 4: 7
Enter number 5: 9
Average = 5.0
○ kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$
```

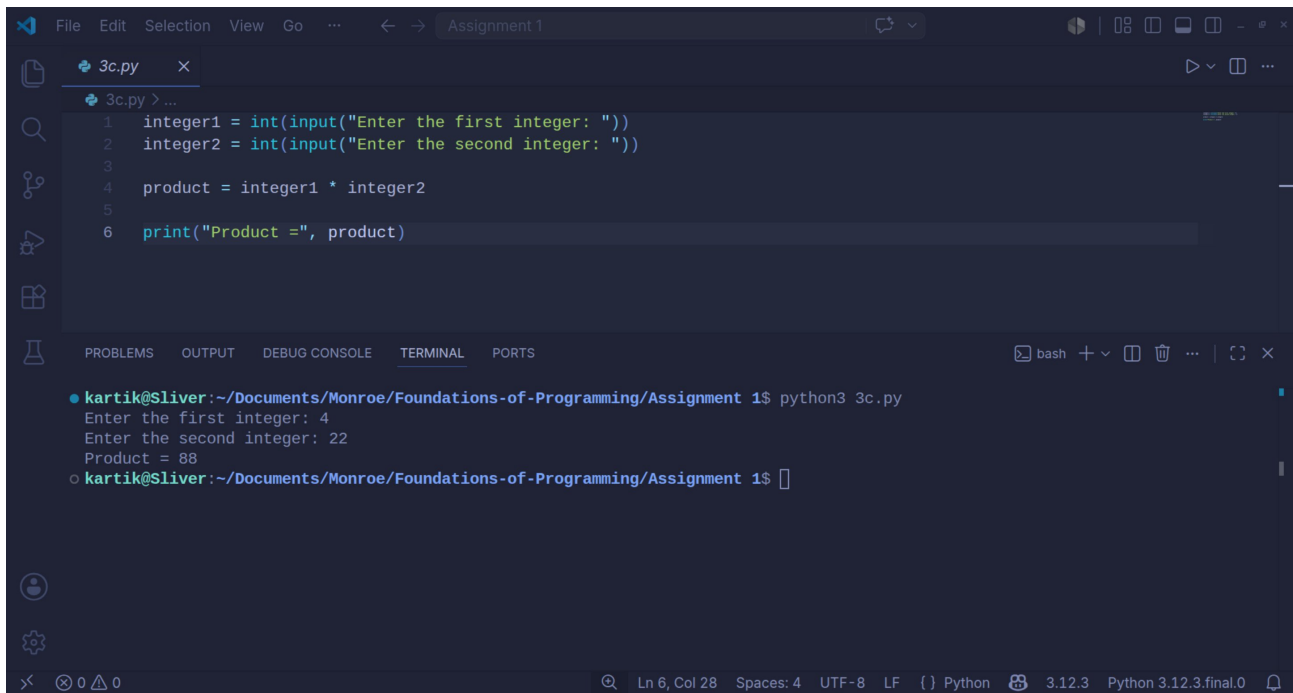
The status bar at the bottom indicates the file is at line 6, column 54, with 4 spaces, using UTF-8 encoding and LF line endings. The Python version is 3.12.3.

c. Multiply Two Integers

```
integer1 = int(input("Enter the first integer: "))
integer2 = int(input("Enter the second integer: "))

product = integer1 * integer2

print("Product =", product)
```



The screenshot shows a code editor with a file named `3c.py` containing the following Python code:

```
1 integer1 = int(input("Enter the first integer: "))
2 integer2 = int(input("Enter the second integer: "))
3
4 product = integer1 * integer2
5
6 print("Product =", product)
```

Below the code editor is a terminal window. The terminal shows the command `python3 3c.py` being executed. The output of the program is:

```
kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$ python3 3c.py
Enter the first integer: 4
Enter the second integer: 22
Product = 88
kartik@Sliver:~/Documents/Monroe/Foundations-of-Programming/Assignment 1$
```

The terminal window also shows the file path `~/Documents/Monroe/Foundations-of-Programming/Assignment 1` and the command `python3 3c.py`.